

Overprecision and (Ir)rational Inattention ^{*}

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Abstract

This paper integrates overprecision—a form of overconfidence where individuals overestimate the accuracy of their beliefs—into a canonical rational inattention model. We show that overprecision distorts belief updating directly by biasing the perceived value of new information and indirectly by amplifying the impact of attention costs. We test the model’s predictions in a pre-registered 2 x 2 belief-updating experiment that manipulates overprecision and information costs. The results confirm that overprecision reduces updating and that higher information costs lead to lower responsiveness to signals. While our pre-registered analysis finds no support for the predicted interaction between overprecision and attention costs, a more granular specification shows that the effect of information costs on belief updating is stronger for overprecise participants. These findings suggest that what appears as rational inattention partly reflects *irrational* inattention arising from misperceived prior accuracy, underscoring the need to distinguish informational frictions from cognitive biases when modeling attention.

Keywords Overconfidence, Rational inattention, Belief updating, Overprecision

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1 Introduction

The canonical full information rational expectations (FIRE) framework assumes that agents form beliefs by optimally processing all available information. However, empirical evidence often contradicts this assumption.¹ A prominent explanation for these departures from the Bayesian benchmark is "rational inattention" (Sims, 2003). At the core of this theory is the premise that processing information is costly. Therefore, if the value of processing a unit of information does not exceed its cost, then ignoring that information becomes the "rational" course of action. This theory has emerged as a key framework, not only because it can explain a wide variety of economic phenomena such as financial contagion (Mondria and Quintana-Domeque, 2012) or the slow adjustment of prices to nominal shocks (Woodford, 2001; Sims, 2003), but also because it appeals to a common human behavior—ignoring information.²

While the assumption that agents can assess the costs and benefits of acquiring and processing information without bias serves as a useful benchmark, it is unlikely to hold in practice. A large body of literature has shown that individuals deviate significantly from rational or fully Bayesian optimization due to biases such as base rate neglect (Kahneman and Tversky, 1973; Tversky and Kahneman, 1974), correlation neglect (Enke and Zimmermann, 2019), environmental complexity (Oprea, 2020), or simply limited cognitive ability (D'Acunto et al., 2021). More recently, the literature has highlighted overprecision as a systematic and pervasive bias shaping various aspects of decision-making and behavior (Kahneman, 2011; Moore et al., 2015; Bosch-Rosa et al., 2024).

Overprecision is a form of overconfidence arising from a metacognitive bias whereby individuals systematically overestimate the precision of their information (Moore, 2022). Relative to Bayesian agents with calibrated beliefs about their uncertainty, overprecise agents are *wrong* about the precision of their prior knowledge. Consequently, the perceived reliability of beliefs is inflated, biasing the cost-benefit evaluations of trade-offs in information processing and attention allocation.³

¹While some studies show underreaction to changes in taxation (Chetty et al., 2009; Taubinsky and Rees-Jones, 2018) or to important monetary policy announcements (Coibion et al., 2021, 2020), others document overreaction to macroeconomic or financial news by households, firms, and experts (Bordalo et al., 2020; Broer and Kohlhas, 2024). See Born et al. (2023) for a recent literature review on the FIRE.

²See Maćkowiak et al. (2023) for a recent overview of the rational inattention literature.

³The negative implications of such miscalculation have been documented across a wide range of do-

This paper examines how overprecision influences (in)attention and shapes individuals’ information use, theoretically and empirically. Specifically, we show that overprecision may lead to systematic deviations from optimal attention allocation by distorting perceived cost-benefit trade-offs in information processing. Our key insight is that overprecision introduces a two-layered distortion. First, it biases the perceived precision of the agents’ prior beliefs, leading them to overestimate the extent of their current knowledge. Second, because agents see fewer gains from processing new signals when they think their baseline knowledge is highly precise, they may exhibit “irrational inattention,” allocating less attention than standard rational inattention models would predict.

We test our model’s predictions using a novel task embedded in a pre-registered, incentivized 2 x 2 online experiment conducted on a large, representative German sample. The survey consists of two blocks. In the first block, participants view a series of pictures depicting groups of people and are asked to estimate the average age of each group at the time the picture was taken. For each picture, we also elicit participants’ expected *absolute* error.⁴ Following the “Subjective Error Method” (SEM) introduced in Bosch-Rosa et al. (2024), we measure overprecision as the difference between expected and realized absolute errors across Block 1. In Block 2, participants again provide average age estimates and expected absolute errors for a new set of pictures. However, in this block, participants can update their estimates for each picture after viewing an “information cloud” containing a random subset of ages from the picture.⁵

To identify the causal effects of overprecision and inattention on belief updating, we implement two experimental treatments designed to independently manipulate these components in Block 2. For overprecision, a randomly selected subset of participants receives calibration feedback indicating whether their absolute error estimates in Block 1 were

mains, such as excessive trading in the stock market (Barber and Odean, 2001), the formation of systematic forecasting errors in finance (Deaves et al., 2019), political extremism (Ortoleva and Snowberg, 2015), or the prevalence of “fake news” (Thaler, 2024).

⁴As an example, imagine we asked about the number of teeth an adult polar bear has. While participants will try to provide their best answer (with an *expected* error of zero), they will also be aware that their answer is most likely incorrect (Yeung and Summerfield, 2012; Bosch-Rosa et al., 2024). Their anticipated deviation from the true value, regardless of direction, is what we try to capture by asking for the absolute error.

⁵Because in what follows the state is *continuous* (the outcome is an average age), which lets us separate mean and variance of beliefs. Accordingly, “overprecision” means an *overly narrow* prior (perceived prior variance too low), whereas “underprecision” means an *overly wide* prior (perceived prior variance too high).

overprecise or underprecise. The control group receives no feedback. By comparing pre- and post-feedback blocks, we can identify the causal effect of calibration feedback on overprecision. For inattention, we randomly assign a subset of participants to a treatment that makes information processing harder by including irrelevant “decoy” words—such as “table” or “spaghetti”—into the information cloud, effectively increasing information complexity without altering the signal’s precision. Together, these features let us study updating conditional on measured overprecision and test how overprecision amplifies the effect of higher attention costs.

Our experimental design includes several unique features. First, although prior research has examined the effects of calibration feedback on overconfidence (e.g., [Russo et al., 1992](#); [Moore and Schatz, 2020](#)), these studies measure overprecision using confidence intervals, a method known to be problematic due to respondents’ unfamiliarity with them ([Teigen and Jørgensen, 2005](#); [Moore et al., 2015](#)). In contrast, we measure overprecision using the SEM, a method that is robust across knowledge domains and heterogeneous populations ([Bosch-Rosa et al., 2024](#)).⁶ Second, we employ decoy words as a simple yet novel treatment to manipulate attention costs. To our knowledge, the closest examples are [Ambuehl et al. \(2025\)](#) and [Bronchetti et al. \(2023\)](#); where participants are presented with two-digit addition problems containing both correct and incorrect answers, and are asked to assess the likelihood that at least half are correct. Third, we use a continuous state variable (age) to distinguish between the mean and variance of beliefs. Unlike the “colored dot” paradigm used in previous attention cost experiments (e.g. [Dewan and Neligh, 2020](#)), this more naturalistic measure induces variation in priors and, consequently, overprecision.

Our results confirm the effectiveness of our experimental treatments. Feedback reduced overprecision among participants who were overprecise in the first picture block, while increasing it among those who were underprecise. The attention cost treatment also worked as expected: respondents in the “decoy” condition revised their beliefs less, made larger estimation errors, and spent more time studying the information cloud. Regarding belief updating, participants who received feedback indicating they were overprecise placed greater weight on the signal in the second block. In other words, reducing over-

⁶Another line of literature considers the effect of measurement scales on the amount of measured overprecision ([Haran et al., 2010](#)). Our method avoids this issue by not requiring scales, thereby circumventing the pitfalls associated with this dimension.

precision led participants to view their prior beliefs as less accurate, thereby shifting the perceived cost-benefit analysis in favor of allocating more attention to new information.

Our pre-registered analysis does not detect the predicted interaction between overprecision and attention costs. However, exploratory analysis suggests that the interaction null result is likely due to feedback influencing both overprecision and perceived task difficulty, introducing a confound that masks the interaction effect in aggregate comparisons. To address this, we introduce a more granular statistical model that isolates the share of belief updating attributable to new information. The results indicate that overprecise participants are systematically less responsive to signals when processing costs are high and make larger estimation errors in the “decoy” condition. Together, these findings support the model’s prediction that *irrational inattention* emerges from the interaction between biased priors and increased information complexity.

Our study contributes to the growing literature on rational inattention by showing how cognitive biases can interact with information-processing costs to generate systematic deviations from optimal attention allocation. As noted by [Maćkowiak et al. \(2023\)](#), an important direction for future research is to extend the rational inattention framework beyond its current application in explaining behavioral anomalies to examine how these anomalies interact with information frictions. In line with this call, we show that overprecision, a well-documented cognitive bias, distorts attention allocation by altering the perceived value of new information. This contributes to a broader shift in the literature, from treating inattention as purely rational to recognizing how miscalibrated priors or distorted mental models can drive it. We also contribute to the literature on how non-standard preferences and beliefs shape attention, such as [Pagel \(2018\)](#), who incorporates prospect theory into portfolio choice under inattention, and [Bolte and Raymond \(2024\)](#), who show that payoff-dependent emotions influence attention allocation. In contrast to these preference-driven mechanisms, our results identify a belief-based distortion: overprecision inflates perceived prior accuracy, thereby reducing responsiveness to costly information. This mechanism is consistent with the framework proposed by [Gabaix \(2019\)](#), in which behavioral phenomena like money illusion and base-rate neglect are driven by internally biased information processing. We extend this perspective by isolating overprecision as a specific cognitive distortion that amplifies attentional frictions and leads to suboptimal belief updating.

We also contribute to the literature on overprecision, a pervasive bias linked to a variety of negative outcomes and described by [Bazerman and Moore \(2012\)](#) as “the mother of all biases.” Precisely because of this relevance, trying to reduce overprecision through feedback has been an active area of research ([Moore and Healy, 2008](#); [Haran et al., 2010](#)), though such interventions have proven notoriously difficult ([Moore et al., 2015](#); [Sanchez and Dunning, 2023](#)). In contrast, we show that if measured using the *subjective error method* ([Bosch-Rosa et al., 2024](#)), feedback can reduce overprecision in a simple context in the short run. More importantly, our results show that overprecision can lead to a misallocation of attention, making people “irrationally inattentive.”

Irrational inattention is particularly problematic because, unlike standard rational inattention, it does not respond predictably to incentives. While increasing the rewards for accuracy can offset attention costs under rational inattention, our model predicts that overprecision continues to suppress belief updating even when the stakes are high. This prediction is supported by our empirical finding that overprecision reduces responsiveness even when information is easy to process. Beyond validating the model, this distinction has direct practical implications for the design of information policies and behavioral interventions. For example, in economic policy contexts, raising the salience of monetary or fiscal announcements may improve responsiveness among inattentive individuals, but may have limited impact on those exhibiting overprecision. In settings where public information uptake is crucial (e.g., health advisories, financial education, or civic communication), interventions that target biased confidence in prior beliefs may be necessary to promote more accurate belief updating and improved decision-making.

2 Model & Hypotheses

We start with a standard Bayesian updating framework where agent i has a prior belief about the uncertain fundamental θ . These beliefs are normally distributed with mean $\mu_{\theta,i}$ and variance $\sigma_{\theta,i}^2$. The agent uses the perceived variance $\tilde{\sigma}_{\theta,i}^2$ rather than objective variance $\sigma_{\theta,i}^2$ with the difference depending on the agents’ overprecision ω_i such that

$$\tilde{\sigma}_{\theta,i}^2 = \frac{\sigma_{\theta,i}^2}{\omega_i}$$

with $\omega_i \in (0, \infty)$. Therefore, if an agent is overprecise ($\omega_i > 1$), it perceives its prior belief as *more* precise than it really is. If the agent is underprecise ($\omega_i < 1$), it perceives its prior belief as *less* precise than it really is.

Agents receive an unbiased noisy public signal x about the fundamental:

$$x = \theta + \epsilon,$$

where $\epsilon \sim N(0, \sigma_\epsilon^2)$. The signal noise is orthogonal to the prior belief noise. We follow the convention in the rational inattention literature and model inattention with an individual-specific noise term ψ_i (e.g. Fuster et al., 2022; Sims, 2003). Therefore, agent i perceives the signal as

$$s_i = x + \psi_i,$$

where ψ_i is assumed to be normally distributed as $N(0, \sigma_{\psi,i}^2)$. The variance of this noise symbolizes the level of inattention.

In this setup, the posterior belief of agent i can be written as the weighted average of the signal and the prior mean:

$$E[\theta \mid s_i] = \beta_i \cdot (\theta + \epsilon + \psi_i) + (1 - \beta_i) \cdot \mu_{\theta,i}, \quad (1)$$

where the weight on the signal (i.e., the update rate) can be expressed as the ratio of the variances

$$\beta_i = \frac{\tilde{\sigma}_{\theta,i}^2}{\tilde{\sigma}_{\theta,i}^2 + \sigma_\epsilon^2 + \sigma_{\psi,i}^2}. \quad (2)$$

Therefore, overprecision skews agents' weighting of new information.

2.1 Optimal Updating

Agents determine the optimal level of attention (i.e., $\sigma_{\psi,i}^{-2}$) by trading off posterior accuracy against attention costs. More formally, they solve:

$$\max_{\sigma_{\psi,i}^{-2}} -\phi_i \underbrace{E[(\theta - E[\theta \mid s_i])^2 \mid s_i]}_{=Var(\theta|s_i)} - C(\sigma_{\psi,i}^{-2}). \quad (3)$$

The first term refers to the expected quadratic belief error for the level of processed signal.⁷ The scalar ϕ_i is a parameter capturing individual-specific incentive to form more accurate beliefs, while the second term ($C(\sigma_{\psi,i}^{-2})$) quantifies the cost of attention. We assume this cost is proportional to the expected reduction in uncertainty, measured by Shannon entropy:

$$C(\sigma_{\psi,i}^{-2}) = g(H(x) - H(x | s_i)).$$

Here the difference $H(x) - H(x | s_i)$ is the mutual information which measures how much uncertainty (or surprise) the agent eliminates by processing the signal (or paying attention). Given that the entropy of a Gaussian random variable is $H(x) = c + \frac{1}{2} \ln(\sigma_x^2)$ —where c is a constant—we can express the cost function as:

$$C(\sigma_{\psi,i}^{-2}) = g\left(\frac{1}{2} \ln\left(\frac{\sigma_x^2}{\sigma_{x|s}^2}\right)\right) = g\left(\frac{1}{2} \ln\left(1 + \frac{\tilde{\sigma}_{\theta,i}^2 + \sigma_\epsilon^2}{\sigma_{\psi,i}^2}\right)\right). \quad (4)$$

A common approximation is to assume that the function g is linearly related to the expected reduction in uncertainty such that

$$C(\sigma_{\psi,i}^{-2}) = \frac{\lambda_i}{2} \ln\left(1 + \frac{\tilde{\sigma}_{\theta,i}^2 + \sigma_\epsilon^2}{\sigma_{\psi,i}^2}\right) \quad (5)$$

In this case, the parameter λ_i reflects the marginal cost of attention.

By substituting equation (5) into equation (3) and using the expression in (2), we can rewrite the maximization problem as a function of the update rate β_i :

$$\max_{\beta_i} -\phi_i(1 - \beta_i)\tilde{\sigma}_{\theta,i}^2 - \frac{\lambda_i}{2} \ln\left(\frac{1}{1 - \beta_i \frac{\tilde{\sigma}_{\theta,i}^2 + \sigma_\epsilon^2}{\tilde{\sigma}_{\theta,i}^2}}\right).$$

Solving this problem for β_i yields the optimal solution:

$$\beta_i^* = \max\left\{0, \frac{\tilde{\sigma}_{\theta,i}^2}{\tilde{\sigma}_{\theta,i}^2 + \sigma_\epsilon^2} - \frac{\lambda_i}{2\phi_i\tilde{\sigma}_{\theta,i}^2}\right\}. \quad (6)$$

The two-layered effect of overprecision on posterior beliefs is apparent in equation (6):

⁷Because the Bayesian point forecast is unbiased, the conditional mean-squared error equals the posterior variance.

the first term represents the standard Bayesian update rate, adjusted by the prior variance, which is biased by overprecision, while the second term captures the effect of inattention, which is also influenced by overprecision.⁸

2.2 Hypotheses

Assuming equation (6) has an interior solution, we can derive several hypotheses. The first two have to do with the direct effects of overprecision and attention costs:

Hypothesis 1: Increasing overprecision leads to a lower update rate:

$$\frac{\partial \beta_i^*}{\partial \omega_i} = -\frac{\sigma_\epsilon^2 / \sigma_{\theta,i}^2}{(1 + \omega_i \sigma_\epsilon^2 / \sigma_{\theta,i}^2)^2} - \frac{\lambda_i}{2\phi_i \sigma_{\theta,i}^2} < 0$$

Hypothesis 2: Increasing attention cost leads to a lower update rate:

$$\frac{\partial \beta_i^*}{\partial \lambda_i} = -\frac{\omega_i}{2\phi_i \cdot \sigma_{\theta,i}^2} < 0$$

These hypotheses suggest an underreaction to information compared to the FIRE benchmark. However, the mechanisms differ. Hypothesis 1 captures *irrational inattention* as an underreaction from biased priors due to overly narrow priors. In contrast, Hypothesis 2 reflects rational inattention, where underreaction is optimal as agents deliberately limit information processing due to the explicit attention costs. A key implication is that while increasing incentives ϕ_i may fully eliminate underreaction due to attention costs, they are insufficient to offset the effects of overprecision.

The third hypothesis has to do with the interaction between overprecision and attention costs:

Hypothesis 3: Overprecision amplifies the negative effects of attention costs on the update rate:

$$\frac{\partial}{\partial \omega_i} \left(\frac{\partial \beta_i^*}{\partial \lambda_i} \right) = -\frac{1}{2\phi_i \sigma_{\theta,i}^2} < 0.$$

Because $\frac{\partial \beta_i^*}{\partial \lambda_i} = -\frac{\omega_i}{2\phi_i \sigma_{\theta,i}^2}$, the slope of β_i^* in λ_i is more negative at higher ω_i ; taking $\partial/\partial \omega_i$ yields the constant $-\frac{1}{2\phi_i \sigma_{\theta,i}^2} < 0$, i.e., higher overprecision mechanically magnifies the impact of attention costs. This interaction captures the indirect channel through

⁸Note that in this formulation of equation (6), ϕ_i and λ_i are non-negative to ensure that each term's contribution to the update rate β_i^* is non-positive when considered independently.

which overprecision influences belief formation via the misallocation of attention. Figure 1 illustrates this effect by plotting the optimal update rate (vertical axis) as a function of overprecision (horizontal axis) and varying levels of attention cost (depicted by different line types) for the same level of objective prior variance and signal precision. As can be seen, the more overprecise one becomes, the greater the effect of changing the cost of attention, up to extreme levels where the agent does not update at all.

In Figure 1, we visualize the interaction captured in Hypothesis 3. The vertical axis reports the optimal updating weight β_i^* , the horizontal axis is overprecision (ω), and each curve corresponds to a different attention cost (λ). Comparing vertical moves $A \rightarrow A^*$ at $\omega = 2$ and $B \rightarrow B^*$ at $\omega = 1.5$, each distance is the loss in β_i^* when the attention cost rises from $\lambda = 1$ (dotted) to $\lambda = 2$ (dashed). Since $AA^* > BB^*$, the *same* increase in cost depresses updating *more* the more overprecise an agent is.

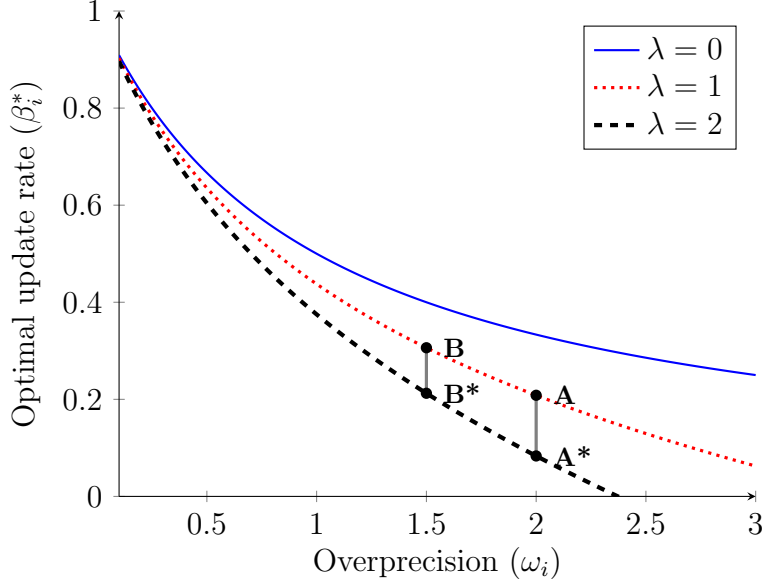
This interaction has concrete behavioral implications. When costs tick up, overprecise agents disproportionately forgo information, make fewer state-contingent adjustments, and display more inertia in their choices. Their forecasts and actions react less to new evidence, so errors persist longer and convergence to the true state is slower. By contrast, better-calibrated agents still see enough value in the signal to keep attending and updating, so the same cost increase produces a smaller drop in β_i^* . That is exactly what Figure 1 visualizes: the vertical gap between the $\lambda=1$ and $\lambda=2$ curves grows with ω .

2.3 Model Extensions

To verify that our comparative static results are not an artifact of specific modeling assumptions, we consider two different extensions. In the first one, we extend the model to a multi-signal environment in which the agent can process a sequence of independent signals and re-optimize attention after each observation. This multi-stage environment resembles our experiment, in which subjects first see a picture and then receive numerical information about the ages of the people in the picture. Resolving the problem recursively shows that the perceived prior precision carried into each stage largely preserves the same qualitative comparative statics for the stage- t update weight β_t as in the static case. See Appendix D.1 for details.

In the second extension, we relax the specific entropy-based cost function, which has been criticized in the literature (e.g., Dean and Neligh, 2023), and show that the results

Figure 1: Optimal updating by varying overprecision and cost of attention



Calibration: $\phi = 2$, $\sigma_\epsilon^2 = 4$, and $\sigma_{\theta,i}^2 = 4$

do not hinge on this specific form. For a broad class of increasing, convex attention cost functions, such as linear, exponential, or power functions, the predictions remain unchanged under interior solutions. Appendix D.2 contains the derivations.

3 Experimental Design

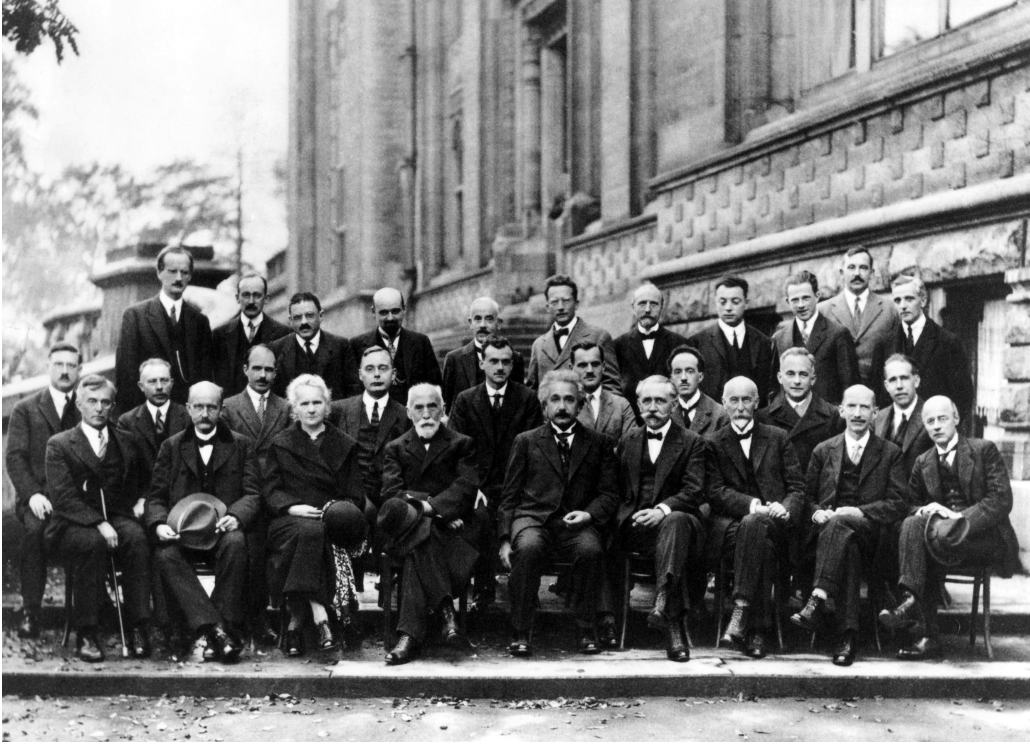
3.1 Design

To test our model’s predictions, we conduct a 2x2 between-subjects online experiment.⁹ The experiment consists of two blocks. In the first block, participants view five historical pictures, such as the one shown in Figure 2 (see Appendix C.1 for the full set).¹⁰ For each picture, participants are asked to estimate the average age of the people in the image at the time the picture was taken. After submitting their estimate, participants are asked to assess the absolute distance between their estimate and the true average age in the picture. That is, participants are asked to estimate their “subjective error” (Bosch-Rosa et al., 2024). Both the age estimate and the subjective error are restricted to a range from 0 to 100.

⁹The experimental design and analysis plan are pre-registered at AsPredicted #154320.

¹⁰The pictures cover a range of topics, including politics, popular culture, and the arts. The order of the pictures for each block was randomized at the participant level.

Figure 2: Example of picture used in the online experiment.



Note: This picture depicts the Fifth Solvay Conference, which was held in October 1927. The average age of the people in the picture is 45.83 years. The youngest persons (Paul Dirac and Werner Heisenberg) are 25 years old, while the oldest person (Hendrik Lorentz) is 74 years old.

At the end of the first block, two-thirds of participants are randomly assigned to the *Feedback* treatment, which provides feedback on the accuracy of their subjective errors. The remaining third are assigned to the *No-Feedback* treatment and receive no feedback. Specifically, subjects in the *Feedback* treatment receive the following text:

“In more than half of the last five pictures, your answer was [*further away from the correct answer*][*closer to the correct answer*] than you expected”

depending on the performance.¹¹ We refer to these as *Overprecise* and *Underprecise* feedback, respectively.

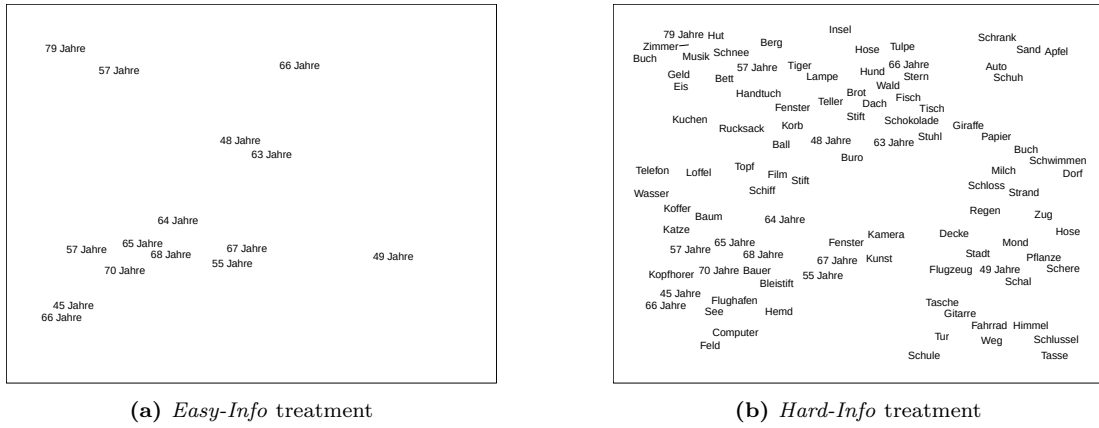
After receiving feedback, participants begin Block 2. In it, participants once again estimate the average age of individuals and report their subjective error for five new pictures. The difference between Block 1 and Block 2 is that, following the belief and

¹¹We oversample the *Feedback* treatment to ensure sufficient statistical power for each type of feedback.

subjective error report, participants are shown an “information cloud” containing fifteen unique ages of people in the picture.¹²

Participants are randomly assigned to one of two informational treatments. In the *Easy-Info* treatment, the cloud displays the fifteen ages. In the *Hard-Info* treatment, the cloud contains the same fifteen ages in the same locations. However, it also includes 85 additional random words (decoys) interspersed within the ages, increasing processing complexity without changing signal precision (see Figure 3 for an example).¹³ After viewing the cloud, participants are asked to enter their updated estimate of the average age in the picture.

Figure 3: Example of an information cloud in the *Easy-Info* and *Hard-Info* treatments



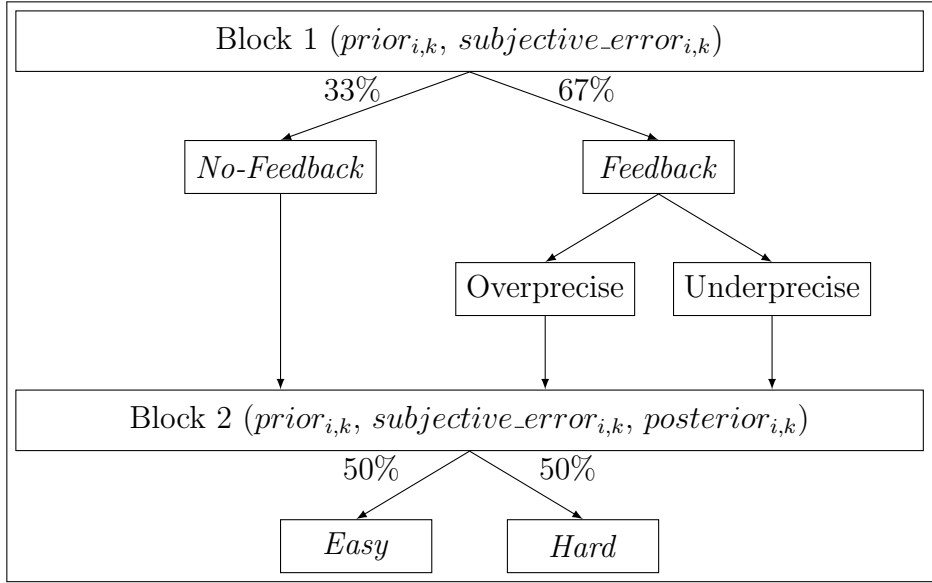
Note: Left panel shows the information cloud containing 15 ages of people in the picture. The right panel shows the format in the *Hard-Info* condition, where the same ages are in identical positions, along with 85 random words. All decoy words are common German nouns; e.g., *Baum* = tree, *Buch* = book, etc. All participants within each treatment saw the same set of randomly drawn ages and words.

A graphical summary of the whole experimental design can be found in Figure 4. The 2x2 experimental design allows us to observe belief updating by comparing posterior beliefs to initial beliefs as a function of overprecision and attention cost.

¹²Participants are informed that each cloud contains fifteen ages corresponding to individuals in the picture.

¹³An alternative approach would have been to allow subjects to endogenously select how many ages to observe for a certain given cost. Instead, we opted to exogenously manipulate attention costs through decoy items in the signal cloud. This strategy offers clean identification of causal effects without introducing strategic behavior or confounding variation in effort or motivation.

Figure 4: Overview of the experimental design



Note: Graphical representation of the experimental design. In both blocks, subjects evaluate five different pictures. The pictures in each block are the same for all subjects, but their order is randomized at the subject level. In Block 1, each participant i stated for each of the five pictures k a belief about the average age in the picture ($prior_{i,k}$) and reported a subjective error ($subjective_error_{i,k}$). In Block 2, each participant stated a belief about the average age, the subjective error, and a posterior belief ($posterior_{i,k}$) after seeing the signal.

3.2 Incentives & Procedures

The survey was fielded by Bilendi, a professional survey provider, on a representative sample of the German population. Participants were informed that a randomly selected 5% sample would be eligible for a €10 bonus payment, in addition to the standard fee paid by Bilendi, contingent on the accuracy of their answers. To prevent hedging, participants were told that the bonus payment would depend on the accuracy of a single, randomly selected estimate. Payoffs were determined using the binarized scoring rule (Hossain and Okui, 2013). Following Danz et al. (2022), the formulation of the scoring rule was not directly presented to participants. Instead, they were told that the probability of winning the bonus increases with the accuracy of their answers.¹⁴ Participants had the option to click on a link to read a detailed, formal description of the payoff mechanism.

¹⁴The specific wording was: “Je näher Ihre Antwort auf diese zufällig ausgewählte Frage an der richtigen Antwort liegt, desto höher ist die Wahrscheinlichkeit, dass der potenzielle Bonus dann tatsächlich ausgezahlt wird (für weitere Informationen hier klicken). Daher liegt es in Ihrem Interesse, auf jede Frage eine möglichst genaue Antwort zu geben.” Which in English translates to: “The closer your answer to this randomly selected question is to the correct answer, the higher the probability that the potential bonus will actually be paid out (click here for more information). Therefore, it is in your interest to give as accurate an answer as possible to each question.”

3.3 Sample & Data Selection

Because Bilendi pays participants based on survey completion rather than accuracy, we screened out participants who failed attention checks or answered fewer than two out of our four instruction comprehension questions correctly. Participants who passed the initial screening and completed the survey were considered “completed” surveys. In total, we collected 1321 completed surveys. Following our pre-registration, we dropped observations where the subjective error exceeded the distance between the prior and $\max(\text{prior}, 100 - \text{prior})$. If this behavior occurred more than once for the same participant, we removed all of their answers from the dataset. We also excluded any observation with an estimated average age above 80 or below 20 years; if this occurred more than once for the same participant, we removed all of that participant’s answers from the dataset. We also excluded any participant identified as a “speeder,” defined as those who completed more than 50% of the prior estimates in less than 25% of the median completion time. From the remaining sample, we retained the first 1,200 participants (our pre-registered sample) and dropped the rest.

The final sample is representative of the adult German population in terms of age (mean of 44.5 years), gender (50% female), education (30% hold a university degree), and region. Table A.1 in Appendix A summarizes the demographic characteristics of the sample compared to census data. The median participant spent 13 minutes completing the survey.

4 Results

4.1 Overview of Data

Table 1 provides an overview of the data. From left to right, the table reports the name we use for each picture (subjects do not see it), the average age of the people in the picture on the day it was taken, the average prior belief, the average error made by participants in their estimation, the average subjective error, and the average posterior belief, where applicable.¹⁵

¹⁵Figure B.1 in Appendix B plots the density of the responses for each picture.

Table 1: Average prior beliefs, errors, subjective errors, and posteriors

Picture description	avg. age	prior	error		post.
			actual	subjective	
(1) Solvay Conference (1927)	45.83	56.09	10.99	5.23	.
		(8.62)	(7.44)	(3.83)	.
(1) Band Aid (1984)	27.89	29.84	4.86	3.99	.
		(5.91)	(4.11)	(3.07)	.
(1) Obama Cabinet (2009)	54.00	56.72	5.58	4.63	.
		(6.26)	(4.19)	(3.48)	.
(1) Paramount 75th Anniversary (1987)	52.32	54.10	7.45	5.25	.
		(8.76)	(5.23)	(4.01)	.
(1) Sino-German Consultations (2023)	57.04	53.07	5.93	4.53	.
		(6.17)	(4.97)	(3.09)	.
(2) G20 Summit (2017)	60.97	57.76	5.37	5.01	60.96
		(6.15)	(4.71)	(4.73)	(4.49)
(2) The Irascibles (1950)	40.20	46.10	7.20	4.75	42.20
		(6.42)	(5.63)	(3.97)	(4.58)
(2) Australian Labor Party (1901)	40.42	47.32	8.79	5.12	42.50
		(8.48)	(7.23)	(3.99)	(5.26)
(2) Merkel Cabinet (2019)	52.05	51.47	4.81	4.86	51.65
		(5.99)	(4.49)	(4.81)	(4.42)
(2) British Royal Family (2007)	48.65	49.32	7.66	5.91	49.95
		(9.47)	(5.97)	(4.66)	(8.30)

Notes: (1) and (2) refer to the block number. The year in parentheses refers to the year the picture was taken. Pictures within blocks are presented in random order. Standard deviations are shown in parentheses below the means.

The table shows that participants are engaged with the task, as the average age estimate is quite accurate.¹⁶ The data also indicates overprecision, as the average absolute

¹⁶No participant provided the exact correct answer (to two decimal places) for any picture, nor did

error exceeds the average subjective error for all pictures in the first block. We also observe that the signal influences belief updating as the average posterior belief is closer to the true average age than the corresponding prior in all but one instance. Finally, the gap between subjective and actual errors narrows in the second block, showing that feedback messages help participants calibrate their subjective error estimates.

4.2 Manipulation Checks

The goal of our experimental design is to exogenously manipulate overprecision and attention costs. In this subsection, we analyze whether our interventions had the desired effect.

Overprecision

Following Bosch-Rosa et al. (2024), we define overprecision for subject i and picture k as:

$$overprecision_{i,k} = |error_{i,k}| - subjective_error_{i,k},$$

where $error_{i,k}$ is the actual error made by subject i in estimating the average age of people in picture k , and $subjective_error_{i,k}$ is their estimated subjective error for that picture. We then employ a Difference-in-Differences approach to assess the effect of *Overprecise* and *Underprecise* feedback. Specifically, we run the following regression

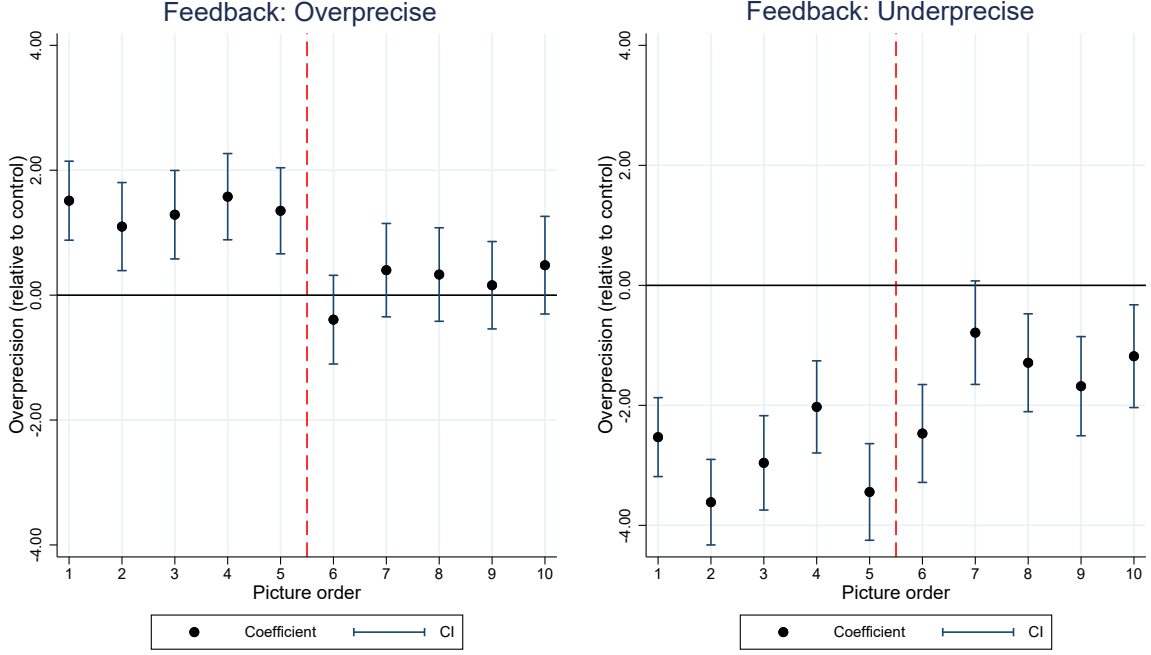
$$overprecision_{i,k} = \alpha + \beta_1 \cdot feedback_i + \beta_2 \cdot picture_order_k + \beta_3 \cdot (feedback_i \cdot picture_order_k) + \epsilon_{i,k} \quad (7)$$

where *feedback* is a dummy variable that takes value one if the participant is in the feedback treatment and *picture_order* is the order of the picture. This regression is estimated separately for overprecise and underprecise participants with OLS, allowing us to compare the feedback-induced overprecision shifts across the two groups.

Figure 5 plots the estimated effect of feedback on overprecision across rounds, along with 90% confidence intervals. It is clear that participants who are initially overprecise

anyone consistently round to the correct answer for all 10 pictures, suggesting that help from search engines or large language models is unlikely.

Figure 5: Overprecision (relative to control) across pictures and feedback treatments



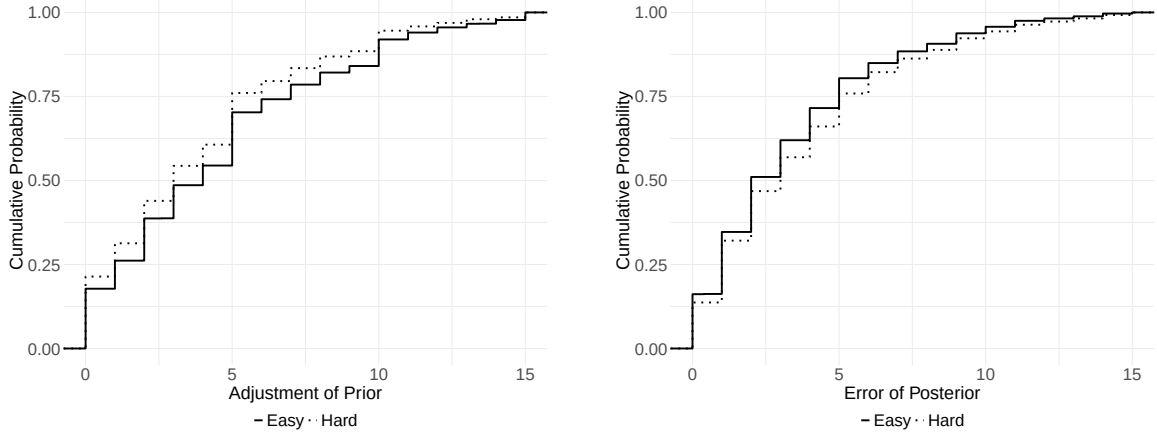
Notes: Estimated overprecision (in years) by picture and feedback treatment indicators. The dependent variable in the regressions is overprecision, and the regressors are indicators for pictures (by order of appearance), feedback treatments, and the interaction between the two. The red dashed line indicates the start of the second block. Bounds are 90% confidence intervals. Standard errors are clustered at the individual level.

become less overprecise after the *Overprecise* feedback, and vice versa for those with *Underprecise* feedback.

Attention Costs

To assess whether decoys in information clouds increase attention costs, we compare the *Easy-Info* and *Hard-Info* treatments using three proxies for information processing costs. The first proxy measures belief adjustment after exposure to the information cloud ($adjustment_{i,k} = |posterior_{i,k} - prior_{i,k}|$). If attention costs are higher in *Hard-Info*, participants should adjust their posterior beliefs less than in *Easy-Info*. The second proxy, measures posterior accuracy ($error_{i,k} = |correct_{i,k} - posterior_{i,k}|$). If attention costs are higher in *Hard-Info*, posterior estimates should be less accurate than in *Easy-Info*. Finally, suppose attention costs are higher in *Hard-Info*. In that case, participants should spend more time processing the information cloud (i.e., $time_cloud_{i,k}$) in *Hard-Info* than in *Easy-Info*.

Figure 6: CDFs of the adjustment of the prior and error of the posterior across treatments



Note: The left panel plots the CDF of the adjustment made to the prior after seeing the signal across the *Easy-Info* (continuous line) and *Hard-Info* treatments (dotted line). The right panel plots the CDF of the error made by participants in their posterior belief in the *Easy-Info* (continuous line) and *Hard-Info* treatments (dotted line). For readability, we limit the horizontal axis of both figures to a maximum of 15 adjustments and error, Figure B.5 in Appendix B plots the figure with the whole support.

Figure 6 plots the cumulative density functions (CDFs) of $adjustment_{i,k}$ (left panel) and $error_{i,k}$ (right panel), separated by treatment. Participants in the *Easy-Info* treatment adjust their beliefs more, as indicated by the weak stochastic dominance of the *Easy-Info* CDF over the *Hard-Info* CDF for $adjustment_{i,k}$. Similarly, the posterior beliefs are more accurate in *Easy-Info*, as the *Hard-Info* CDF weakly stochastically dominates that of *Easy-Info* CDF for $error_{i,k}$. We confirm these visual results by regressing $adjustment_{i,k}$ and $error_{i,k}$ on a treatment indicator with clustered errors at the individual level and picture fixed effects. The estimated coefficient for *Hard-Info* is significant and negative for $adjustment_{i,k}$ ($\hat{\alpha} = -0.734$, $p < 0.01$) and significant and positive for $error_{i,k}$ ($\hat{\alpha} = 0.497$, $p < 0.01$), consistent with higher attention costs under *Hard-Info*.

Interpreting treatment effects on $time_cloud_{i,k}$ is more nuanced, since higher processing costs can lengthen decision times or induce some participants to reduce their attention and disengage. Nevertheless, Table 2 shows that the *Hard-Info* treatment is associated with significantly longer viewing times even after controlling for outliers using median-quantile or Huber-robust regressions. Taken together, the evidence from all three time proxies supports the interpretation that the *Hard-Info* treatment increases attention costs.

Table 2: Decoy signals and the information time

Dep. var.:	<i>time_cloud_{i,k}</i>		
	(1)	(2)	(3)
	OLS	Median	Robust
<i>Hard-Info</i>	0.716	1.426***	1.286***
	(1.144)	(0.260)	(0.192)
constant	18.078***	10.151***	10.451***
	(1.116)	(0.318)	(0.235)
Picture fixed effects	Yes	Yes	Yes
<i>N</i>	6000	6000	6000

Notes: The table reports results from regressions in which time spent looking at information is regressed on a treatment indicator and picture fixed effects. Columns (1) to (3) report OLS estimates, median quantile estimates, and robust regression (with Huber weights) estimates. Standard errors are clustered at the individual level for (1). Asterisks indicate results of two-tailed t-tests: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3 Treatment Effects on Belief Updating

Having shown that both treatments work as intended, we study their effects on the updating behavior of participants following the pre-registered regression:

$$\begin{aligned}
posterior_{i,k} - prior_{i,k} = & \alpha + (signal_k - prior_{i,k}) \cdot \\
& (\beta_0 + \beta_1 \cdot \text{Overprecise} + \beta_2 \cdot \text{Underprecise} + \beta_3 \cdot \text{Hard-Info} + \\
& \beta_4 \cdot \text{Overprecise} \cdot \text{Hard-Info} + \beta_5 \cdot \text{Underprecise} \cdot \text{Hard-Info}) + \\
& \theta_j \cdot picture_k + \varepsilon_{i,k}. \quad (8)
\end{aligned}$$

In this regression, participants' belief revision (the dependent variable) is modeled as a function of the difference between the signal (i.e., the average age in the information cloud) and the prior, the treatment combination, and picture fixed effects.

Equation (8) is motivated by our theoretical model, in which a Bayesian agent updates

beliefs as a weighted average of the prior and the signal:

$$\begin{aligned} posterior_{i,k} &= \beta_{i,k} \cdot signal_k + (1 - \beta_{i,k}) \cdot prior_{i,k}. \\ posterior_{i,k} - prior_{i,k} &= \beta_{i,k} \cdot (signal_k - prior_{i,k}). \end{aligned}$$

So we can interpret the sum of $\widehat{\beta}_T$ as the update rate for each given treatment combination $T \in [0, 5]$.¹⁷

Table 3 reports the estimates of equation (8). Columns (1) and (2), labeled “Full Control group”, include all respondents, while columns (3) and (4), labeled “Matched Control Group”, restrict the control sample to subjects who would have received the *Overprecise* or *Underprecise* feedback, respectively.¹⁸ Irrespective of the comparison group, the results align with those of Table 1. First, in the benchmark conditions (*No-Feedback* and *Easy-Info* treatments), participants place more weight on the signal than on their prior belief, as indicated by $\widehat{\beta}_0 > 0.5$. Second, participants who receive *Overprecise* feedback update their beliefs significantly more than those without feedback ($\widehat{\beta}_1 > 0$). By contrast, while *Underprecise* feedback is associated with a lower signal weight ($\widehat{\beta}_2 < 0$), the coefficient is not statistically different from zero.¹⁹ Third, increasing the cost of attention reduces the amount of belief updating ($\widehat{\beta}_3 < 0$). Taken together, these results form the basis of the first set of results:

Result 1: Reducing overprecision increases the update rate, consistent with Hypothesis 1.

Result 2: Increasing the cost of attention decreases the update rate, consistent with Hypothesis 2.

¹⁷For example, the update rate for the combination of *Hard-Info* and *Overprecise* feedback treatments is given by $\widehat{\beta} = \widehat{\beta}_0 + \widehat{\beta}_1 + \widehat{\beta}_3 + \widehat{\beta}_4$.

¹⁸Since we pre-registered directional hypotheses, we report one-tailed t-tests.

¹⁹One possible explanation could be the lower prevalence of underprecision in the sample (27%). Alternatively, many respondents react more strongly to being told they are overprecise than to being told they are underprecise.

Table 3: Belief updating by treatments

	Full Control Group		Matched Control Group	
	<i>Revision</i>	<i>Revision</i>	<i>Revision</i>	<i>Revision</i>
$\beta_0 : \text{signal} - \text{prior}$	0.595*** (0.028)	0.798*** (0.245)	0.584*** (0.037)	0.624*** (0.038)
$\beta_1 : \text{Overprecise} \times (\text{signal} - \text{prior})$	0.065** (0.037)	0.075** (0.033)	0.079** (0.044)	
$\beta_2 : \text{Underprecise} \times (\text{signal} - \text{prior})$	-0.012 (0.041)	-0.006 (0.036)		-0.045 (0.048)
$\beta_3 : \text{Hard-Info} \times (\text{signal} - \text{prior})$	-0.138*** (0.041)	-0.130*** (0.034)	-0.119** (0.052)	-0.176*** (0.064)
$\beta_4 : \text{Overprecise} \times \text{Hard-Info} \times (\text{signal} - \text{prior})$	-0.011 (0.056)	-0.018 (0.046)	-0.030 (0.064)	
$\beta_5 : \text{Underprecise} \times \text{Hard-Info} \times (\text{signal} - \text{prior})$	0.080 (0.061)	0.079 (0.051)		0.118 (0.079)
$\alpha : \text{constant}$	1.389*** (0.108)	2.014** (1.079)	1.337*** (0.150)	1.463*** (0.152)
Picture fixed effects	Yes	Yes	Yes	Yes
Covariates	No	Yes	No	No
N	5946	5862	3580	2366
R^2	0.57	0.58	0.57	0.57

Notes: The table reports results from OLS regressions in which belief revision is regressed on the difference between the signal value and the prior, the treatment indicators, and their interaction using all observations. The second column adds demographic covariates for the regression in the first column. The third and fourth columns match the type of respondents in the control group, using only overprecise or underprecise respondents, respectively. Robust standard errors clustered at the individual level are reported. Asterisks indicate the results of one-tailed t-tests: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. $\hat{\beta}_5$ is in the opposite direction of the pre-registered hypothesis, so the one-tailed test is not significant.

Our third hypothesis (Hypothesis 3) predicts that overprecision amplifies the negative impact of attention costs on belief updating. Specifically, we expect an interaction between interventions that change overprecision (ω) and the marginal cost of informa-

tion processing (λ). However, we do not find evidence consistent with this prediction, as neither of the interaction estimates is statistically different from zero.²⁰ Therefore, we conclude that:

Result 3: We do not find evidence that overprecision amplifies the effect of attention costs on update rates when the cost of attention changes.

Finally, our data shows an asymmetry between *Underprecise* and *Overprecise* feedback. As shown in Table 3 and the manipulation checks in Section 4.2, participants are more responsive to *Overprecise* feedback than to *Underprecise* feedback. One likely reason is that *Overprecise* feedback is perceived as more critical or surprising, as it tells participants they were more confident than they should have been, whereas *Underprecise* feedback is more affirming and may feel less consequential. This difference in perceived salience could explain the stronger response to *Overprecise* feedback. Moreover, this pattern is consistent with Haaland et al. (2023), who emphasize that information interventions tend to have stronger effects when individuals’ prior beliefs are more misaligned with the message’s content.²¹ Similarly, our results show that feedback significantly shifts beliefs for participants who were initially overprecise, while it has little impact on those who were underprecise. These two mechanisms—greater salience of *Overprecise* feedback and treatment effect heterogeneity—likely interact, suggesting that our intervention primarily acts by correcting biased priors, with a particularly strong effect when feedback is both surprising and negative.

Result 4: We find evidence that feedback interventions produce an asymmetric impact. *Overprecise* feedback on participants’ calibration reduces their overprecision, whereas *Underprecise* feedback has less influence on increasing it.

5 Discussion

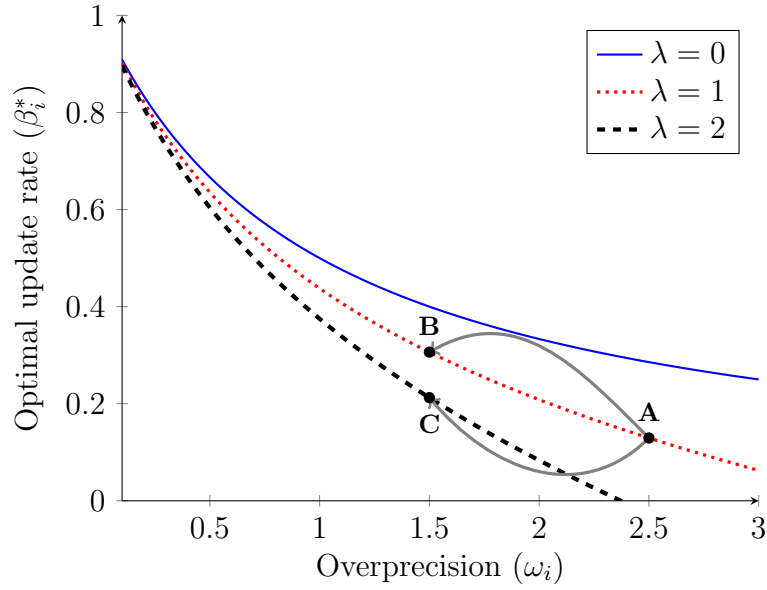
While our results support Hypotheses 1 and 2, the pre-registered specification (eq. (8)) does not support the predicted interaction between overprecision and attention costs

²⁰Although a one-tailed test in the positive direction would yield significance at the 10% level for $\widehat{\beta}_5$, this was not the hypothesized effect.

²¹In our setting, Overprecise feedback challenges participants’ belief in the accuracy of their initial estimates and confidence judgments, while Underprecise feedback largely confirms that participants were cautious or uncertain.

(Hypothesis 3). A plausible explanation is that feedback shifts *both* overprecision and perceived task complexity. In particular, underprecise participants may have been nudged to perceive the task as “simpler” than anticipated (e.g., “This isn’t that hard after all”) while overprecise participants may have been nudged into perceiving the task as “more complex” than anticipated (e.g., “This is trickier than I thought”). This double effect violates *ceteris paribus* and can mask the interaction between overprecision and attention costs.

Figure 7: Optimal updating if feedback treatment had secondary effects on cost of attention



Calibration: $\phi = 2$, $\sigma_\epsilon^2 = 4$, $\sigma_{\theta,i}^2 = 4$.

Figure 7 illustrates this logic. Consider an overprecise participant positioned on the dotted line (point A) who receives feedback in the *Overprecise* treatment. We predict a decrease in overprecision (left shift) and a higher update rate (upward shift), landing at point B. However, the “task-is-harder-than-I-thought” effect may lead the participant to move to a higher perceived cost parameter λ , resulting in a lower update rate (point C). This A-C instead of A-B path can mask the interaction we test in the pre-registered specification.

To empirically evaluate this explanation, we look at the post-experimental survey, where we included a question on the perceived task difficulty. Specifically, we asked participants:

”How difficult was it for you to view (visually) the pictures and word clouds with the information we showed you in the survey?”,

where the responses ($difficulty_i$) are on a Likert scale (1: Not at all difficult - 5: Very difficult).

Table 4 presents the results of regressing $difficulty_i$ on treatment indicators, their interactions, and demographic controls. The estimates show that participants who received Overprecise feedback perceived the task as significantly more complex, suggesting a shift in perceived complexity. Moreover, the absence of interaction effects between the difficulty and Overprecise feedback treatment suggests that feedback alone had a uniform effect across all participants. This broad shift in perceived difficulty may prompt individuals to see everything as more complex, thereby altering their priors, effort, or level of caution in ways that the original difference-based specification of eq. (8) cannot disentangle.

Table 4: Self-reported difficulty of task by treatments

	(1) difficulty	(2) difficulty
<i>Hard-info</i>	0.189* (0.114)	0.191* (0.114)
<i>Overprecise</i>	0.344*** (0.113)	0.346*** (0.114)
<i>Underprecise</i>	-0.023 (0.106)	0.001 (0.107)
<i>Hard-Info</i> $\times$ <i>Overprecise</i>	-0.201 (0.160)	-0.206 (0.161)
<i>Hard-Info</i> $\times$ <i>Underprecise</i>	-0.180 (0.159)	-0.204 (0.160)
Constant	2.059*** (0.077)	2.573*** (0.271)
<i>N</i>	1195	1183
<i>R</i> ²	0.02	0.03
Covariates	No	Yes

Notes: The table reports results from an OLS regression in which self-reported difficulty of the task is regressed on treatment indicators and their interactions. The second column adds demographic covariates for the regression in the first column. Robust standard errors clustered at the individual level are reported. Asterisks indicate the results of two-tailed t-tests: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

To recover the theoretical interaction, we move beyond treatment contrasts and estimate a more granular measure, the “update ratio” ($U_{i,k}$), that directly captures how much participants adjust their beliefs towards the signal. While this analysis was not pre-registered, it offers a more precise test of the mechanism by exploiting within-subject heterogeneity in priors, signals, and responsiveness. The update ratio is defined as:

$$U_{i,k} = \frac{\text{posterior}_{i,k} - \text{prior}_{i,k}}{\text{signal}_k - \text{prior}_{i,k}}. \quad (9)$$

It corresponds to the Bayesian weight on the signal in our model ($\beta_{i,k}$ in eq. 1).²²

The main advantage of focusing on $U_{i,k}$ is that it provides a direct, observation-level measure of responsiveness equal to the fraction of the prior-signal gap closed by the posterior, yielding a scale-free outcome comparable across trials with different prior locations and signal strengths. For example, suppose Overprecise feedback shifts a participant’s priors toward the midpoint of 50 years (perhaps because now the task seems harder). In that case, $U_{i,k}$ still records how much of the remaining gap toward the signal is closed on that trial. This update-ratio approach exploits within-subject variation across trials and, crucially, avoids placing Feedback indicators on the right-hand side (since Feedback affects both calibration and perceived difficulty), yielding a sharper, mechanism-focused test than treatment-level contrasts. Because both $U_{i,k}$ and average overprecision are measured after Feedback, we interpret the estimates as evidence on mechanism rather than causal average treatment effects.

With this, we estimate the model:

$$U_{i,k} = \alpha + \gamma_0 \cdot \text{Hard-Info}_i + \gamma_1 \cdot \text{avg_op}_i + \gamma_2 \cdot (\text{avg_op}_i \cdot \text{Hard-Info}_i) + \gamma_3 \cdot \text{difficulty}_i + \gamma_4 \cdot (\text{difficulty}_i \cdot \text{Hard-Info}_i) + \text{picture}_k + \varepsilon_{i,k}. \quad (10)$$

where avg_op_i is the average overprecision of participant i across the five pictures of block 2, Hard-Info_i is a dummy variable equal to one if the participant was assigned to the Hard-Info condition, and picture_k are picture fixed effects. The coefficients γ_1 and γ_2 capture, respectively, the direct effect of overprecision and its interaction with processing costs. Since $U_{i,k}$ directly proxies for $\beta_{i,k}$, any reduction in the update ratio under high-cost conditions for participants with higher overprecision supports the theoretical amplification effect described in Section 2. Another advantage of this model is that it provides a direct test of whether the “update weight” ($U_{i,k}$) responds to overprecision and to its interaction with processing costs. Because avg_op is measured in Block 2, we include difficulty_i and its interaction with $\text{Hard} - \text{Info}$ on the right-hand side, ensuring that the overprecision terms reflect calibration rather than feedback-induced changes in processing cost.

Table 5 reports the results. In addition to OLS, we report results from quantile and Huber-robust regressions to account for the impact of possible outliers. Across all

²²From eq. 1 we have that $\text{posterior}_{i,k} = \beta_{i,k} \cdot \text{signal}_k + (1 - \beta_{i,k}) \cdot \text{prior}_{i,k}$, which can be rearranged as the update ratio $\beta_{i,k} = \frac{\text{posterior}_{i,k} - \text{prior}_{i,k}}{\text{signal}_k - \text{prior}_{i,k}} := U_{i,k}$.

specifications, we find a robust and statistically significant negative interaction between overprecision and the *Hard-Info* treatment, confirming Hypothesis 3. While the main effect of average overprecision is small and statistically insignificant, this might be because when information is easy to process, even overprecise respondents remain responsive to the signal. It is only when processing costs are high that overprecision meaningfully reduces responsiveness. This result provides evidence for the mechanism of irrational inattention, as overprecision amplifies the effect of information complexity on belief updating. Importantly, the coefficients on perceived difficulty and on its interaction with *Hard-Info* are small and statistically indistinguishable from zero across all three estimators, and including them leaves the overprecision-*Hard-Info* interaction essentially unchanged.

The difference between the pre-registered specification (eq. 8) and the more granular update-ratio is that the former conditions on the prior-signal gap by interacting $(signal_k - prior_{i,k})$ with treatment indicators. Instead, the update-ratio approach places $U_{i,k}$ (i.e., the fraction of the gap closed for each signal) on the left-hand side and tests moderation using continuous overprecision. Because Feedback shifts both calibration and perceived difficulty, including Feedback dummies on the right-hand side can attenuate those interactions. Using $U_{i,k}$ with perceived-difficulty controls (and its interaction with *Hard-Info*) yields a sharper mechanism test and shows that overprecision reduces updating when processing costs are high, but not otherwise.

Table 5: Updating Treatment Interaction

	(1)	(2)	(3)
	update	update	update
<i>avg_op</i>	0.029 (0.025)	0.005 (0.004)	0.002 (0.003)
<i>Hard_info</i>	-0.461* (0.244)	-0.056 (0.054)	-0.046 (0.040)
<i>Hard_info</i> $\times$ <i>avg_op</i>	-0.090** (0.042)	-0.017*** (0.006)	-0.011** (0.005)
<i>difficulty</i>	-0.106 (0.079)	0.004 (0.015)	0.004 (0.011)
<i>Hard_info</i> $\times$ <i>difficulty</i>	0.155 (0.112)	-0.018 (0.021)	-0.014 (0.016)
constant	0.989*** (0.184)	0.746*** (0.045)	0.674*** (0.033)
<i>N</i>	5975	5975	5975

Notes: OLS, quantile regression, and Huber robust regressions of the “update ratio” on the average overprecision of participants (*overprecision_i*), the information treatment (*hard_info_i*), and picture fixed effects (not reported). Errors clustered at the individual level in the OLS. Asterisks indicate the results of two-tailed t-tests: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Finally, as a complementary test of our model, we examine whether the interaction between overprecision and the information costs affects the accuracy of participants’ responses. After all, if reduced updating reflects irrational inattention, it should result not only in lower responsiveness to the signal (as shown in Table 5), but also in less accurate final beliefs. Table 6 reports the results of regressing the absolute posterior error on overprecision, the information treatment dummy, and its interaction. We find a positive and statistically significant interaction between overprecision in the robust specifications (columns (2) and (3)). That is, once we control for extreme values, participants who are both overprecise and assigned to the Hard-Info treatment make systematically larger

mistakes, providing further evidence that overprecision leads to information loss when signal processing is costly, consistent with the lower update ratio shown in Table 5 and with Hypothesis 3.

Table 6: Posterior Absolute Error Interaction

	(1)	(2)	(3)
	error	error	error
<i>avg_op</i>	0.149** (0.065)	0.061** (0.025)	0.069*** (0.017)
<i>Hard_info</i>	0.386 (0.374)	0.140 (0.132)	0.020 (0.090)
<i>Hard_info</i> $\times$ <i>avg_op</i>	0.049 (0.171)	0.183*** (0.035)	0.148*** (0.024)
<i>constant</i>	2.668*** (0.161)	1.933*** (0.146)	2.474*** (0.100)
<i>N</i>	5962	5962	5962

Notes: OLS, quantile regression, and Huber robust regressions of the absolute error in the posterior on the average overprecision of participants (*overprecision_i*), the information treatment (*hard_info_i*), and picture fixed effects (not reported). Errors clustered at the individual level in the OLS. Asterisks indicate the results of two-tailed t-tests: * $p < 0.2$, ** $p < 0.1$, *** $p < 0.02$.

6 Conclusion

This paper identifies overprecision as a behavioral driver of *irrational inattention*. In standard rational inattention, agents optimally trade off information gains against processing costs. In our model, a metacognitive error in which agents *misperceive* the precision of their beliefs distorts both sides of the trade-off making extra information appear less valuable and attention appear less productive.

We test our model’s predictions using a novel 2x2 pre-registered experiment that independently manipulates overprecision via calibration feedback and attention costs via information complexity. This design allows us to provide causal evidence for the impact

of overprecision on belief updating. Specifically, we show that reducing overprecision via feedback increases responsiveness to information while increasing attentional costs diminishes it. However, we do not detect a significant interaction between overprecision and attention costs. In an exploratory analysis, we provide evidence that this null result is likely due to an unintended effect of feedback on participants’ task complexity: informing participants that they were overprecise not only affects their overprecision but also shifts their perceived task complexity, confounding our pre-registered model. To address this, we move beyond binary treatment comparisons and estimate a more granular model that isolates belief updating by directly linking the extent of belief updating toward the signal to participants’ level of overprecision. Using this approach, we find that overprecise participants become especially inattentive when processing information is costly, confirming that subjects exhibit *irrational inattention*.

Overall, our paper contributes both to the theoretical and empirical literature on rational inattention, offering a novel approach to how cognitive biases influence decision-making under information constraints. On the theoretical side, we present a structured framework that integrates overprecision, a well-documented behavioral bias, into a rational inattention model. An important implication of our model is that, while standard inattention can be mitigated by increasing incentives for accuracy, because overprecision is a metacognitive failure, it cannot be corrected by higher stakes alone. This distinction highlights the unique challenge of *irrational inattention* and the negative impact that overprecision can have on individuals’ daily lives (Bosch-Rosa et al., 2024).

Empirically, we provide strong evidence that overprecision can be reduced through feedback. While previous studies have documented the presence of overprecision in various domains (e.g., Moore, 2022; Bosch-Rosa et al., 2024), few have explored whether and how it can be corrected, particularly in contexts involving real-time decision-making under uncertainty. Our results show that targeted feedback effectively recalibrates participants’ priors, reducing overprecision and, consequently, increasing responsiveness to new information. Moreover, the ability to experimentally induce changes in overprecision allows us to identify how biased priors affect attention allocation causally —confirming our theoretical results and demonstrating the presence of *irrational inattention*, a type of inattention driven by biased beliefs rather than rational cost-benefit considerations alone.

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A Additional Tables

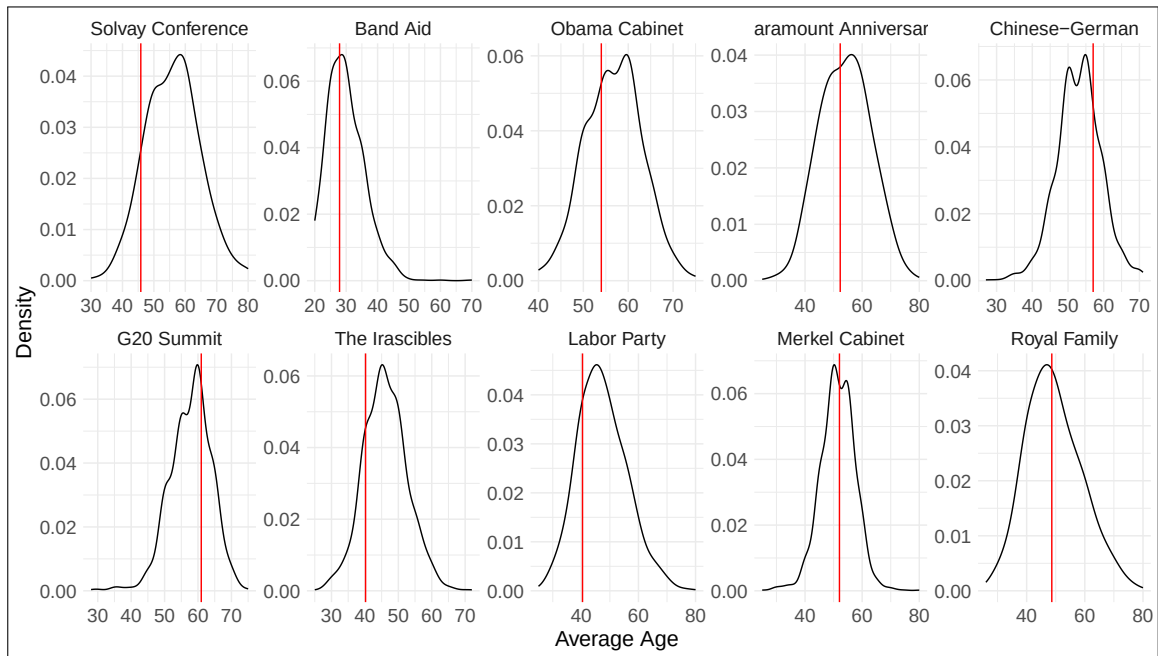
Table A.1: Demographic characteristics of the sample

	Sample	Census [†]
age	44.49 (14.25)	44.6
female	0.504 (0.500)	0.507
German	0.938 (0.242)	0.854
eastern states	0.196 (0.397)	0.194
university degree (or more)	0.312 (0.463) (0.496)	0.335
N	1200	

Notes: East refers to people living in Berlin, Brandenburg, Mecklenburg-Vorpommern, Sachsen, Sachsen-Anhalt, or Thüringen. †: Census data are from Destatis and are as of 2022, except for education statistics, which are as of 2019, and for the adult population ($\text{age} \geq 15$). Standard deviations are shown below the means.

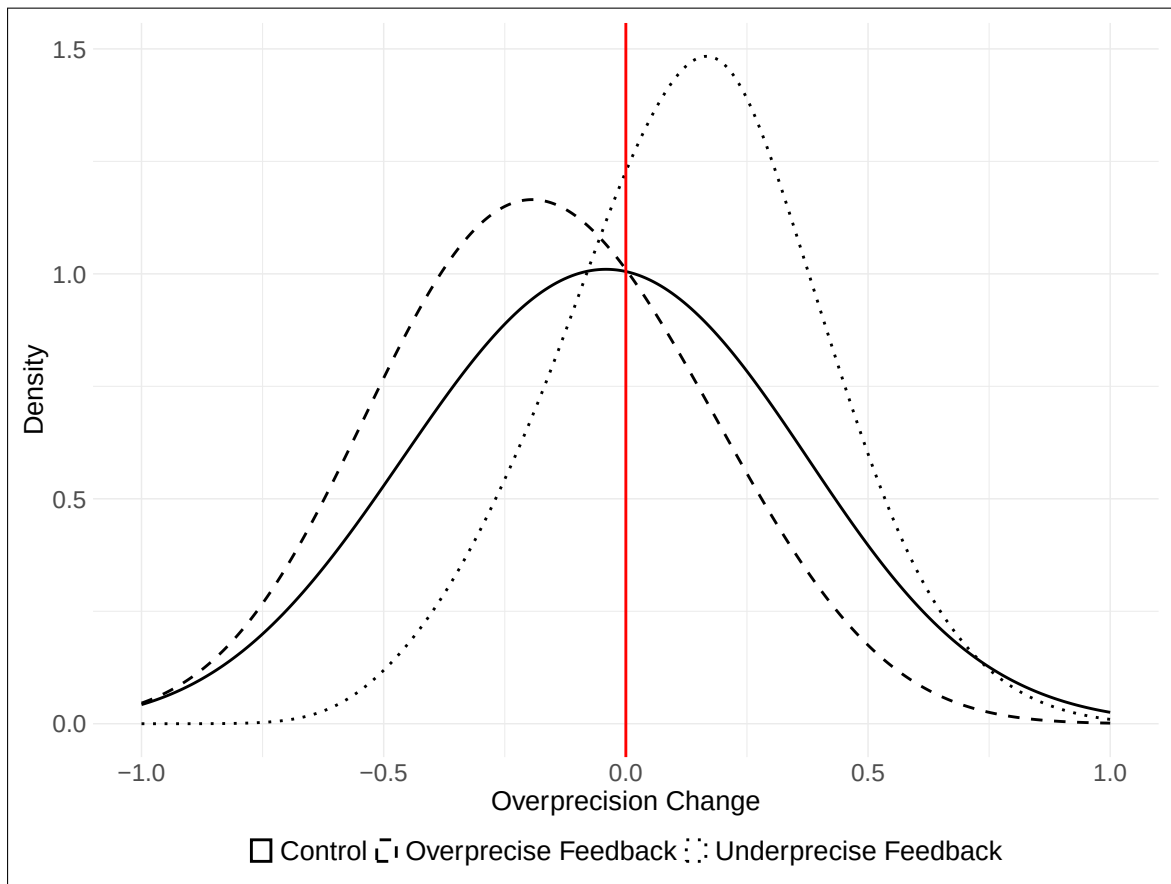
B Additional Figures

Figure B.1: Density of the prior beliefs for each picture



Notes: The vertical red line marks the correct answer. Note that the vertical axis differs for each question.

Figure B.2: Change in overprecision across blocks by feedback type



Notes: The figure shows the density of the difference in average overprecision (binary) between the pictures in the second and first blocks.

Figure B.3: Density of the answers for each picture. The vertical red line marks the correct answer. Note that the vertical axis is the same for all graphs.

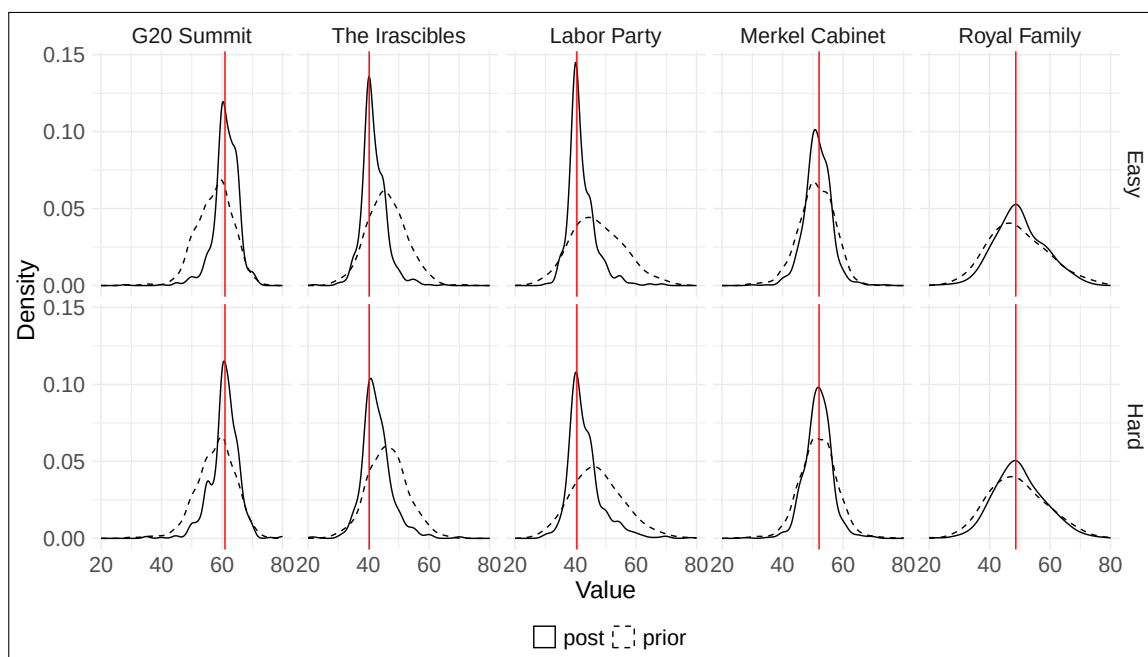


Figure B.4: Density of the answers for each picture. The vertical red line marks the correct answer. Note that the vertical axis differs for each question.

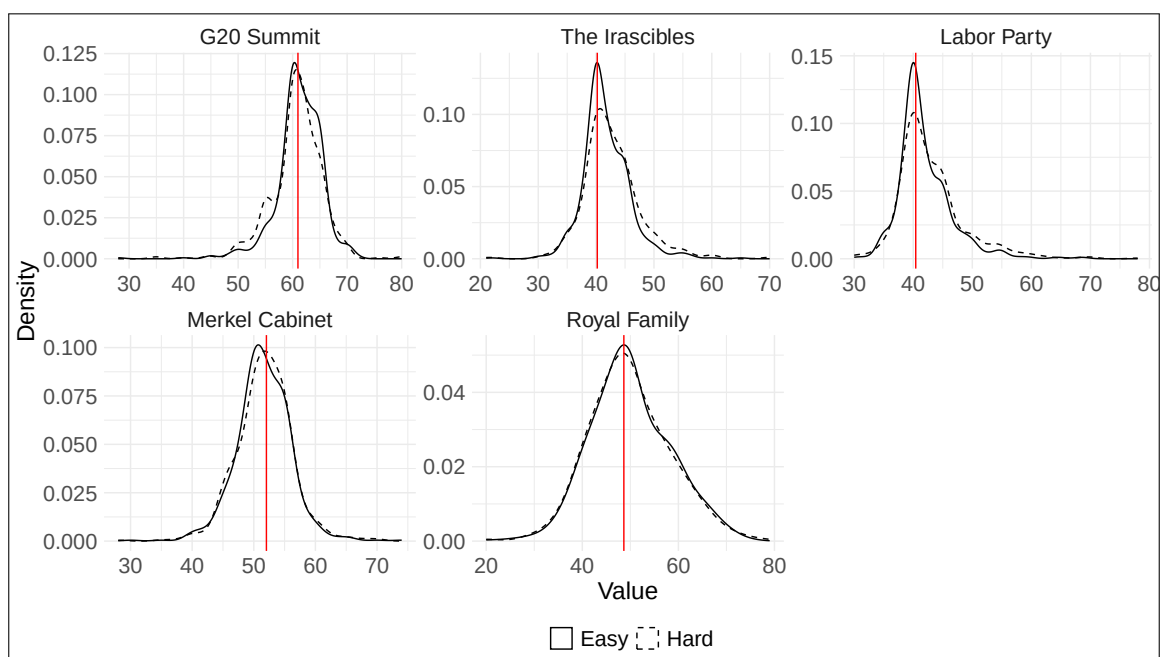
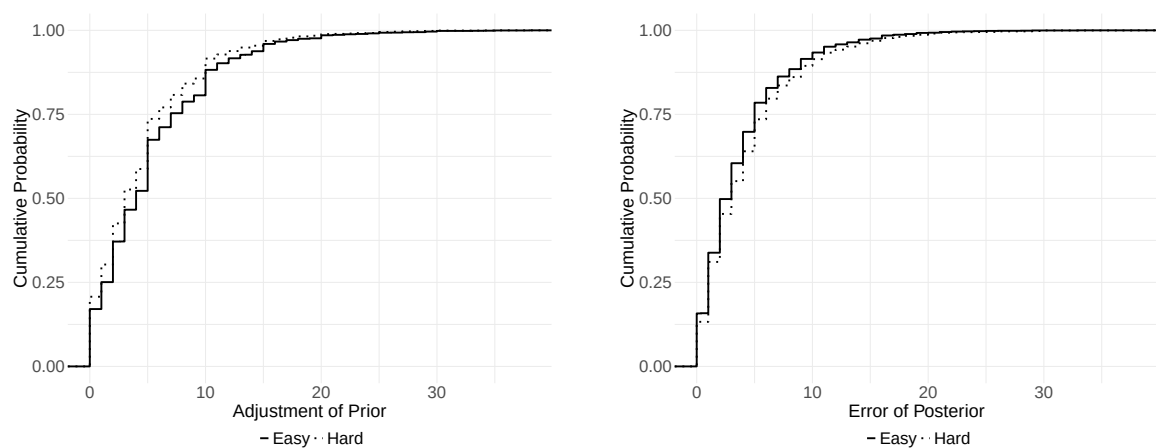


Figure B.5: CDFs of the adjustment of the prior and error of the posterior across treatments



Note: The left panel plots the CDF of the adjustment made to the prior after seeing the signal across the *Easy-Info* (continuous line) and *Hard-Info* treatments (dotted line). The right panel plots the CDF of the error made by participants in their posterior belief in the *Easy-Info* (continuous line) and *Hard-Info* treatments (dotted line).

C Experimental Material

C.1 Pictures

C.1.1 First Block

Figure C.1: The Solvay Conference (1927)

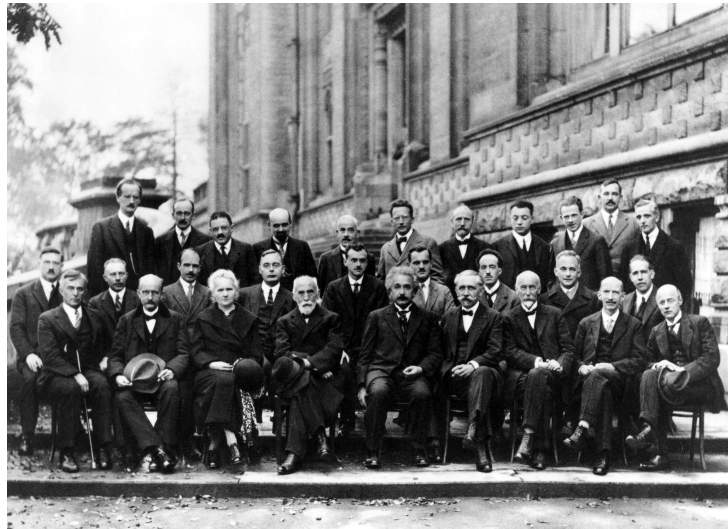


Figure C.2: The Band Aid (1984)



Figure C.3: The Cabinet of the US President B. Obama (2009)



Figure C.4: 75th Anniversary of Paramount (1987)



Figure C.5: The Chinese-German Consultations (2023)



C.1.2 Second Block

Figure C.6: G20 Hamburg Summit (2017)



Figure C.7: The Irascibles (1951)



Nina Lee/Tina Liff Pictures/Getty Images
Life magazine's portrait of the Abstract Expressionist artists known as *The Irascibles*, 1951. Front row: Theodore Stamos, Jimmy Ernst, Barnett Newman, James Brooks, and Mark Rothko; middle row: Richard Pousette-Dart, William Baziotez, Jackson Pollock, Clifford Still, Robert Motherwell, and Bradley Walker Tomlin; back row: Willem de Kooning, Adolph Gottlieb, Ad Reinhardt, and Hedda Sterne

Figure C.8: The Australian Labor Party (1901)



Figure C.9: The Cabinet of German Chancellor Angela Merkel (2019)



Figure C.10: The British Royal Family (2007)



D Model extensions

D.1 Multi-signal environment

This appendix section shows the robustness of the static model predictions to a dynamic setup.

Suppose there are $k = 1, 2, \dots$ independent signals processed sequentially. Before processing the k th signal, the decision-maker chooses the attention level (the precision of idiosyncratic processing noise), denoted $\sigma_{\psi,i}^{-2}(k)$. The k th perceived signal is

$$s_i(k) = \theta + \varepsilon(k) + \psi_i(k), \quad \varepsilon(k) \sim N(0, \sigma_\varepsilon^2(k)), \quad \psi_i(k) \sim N(0, \sigma_{\psi,i}^2(k)), \quad (\text{E.1})$$

with all error terms mutually independent across k and independent of θ . The optimal weight after the first signal coincides with the static case in equation (6):

$$\beta_i^*(1) = \max \left\{ 0, \frac{\tilde{\sigma}_{\theta,i}^2}{\tilde{\sigma}_{\theta,i}^2 + \sigma_\varepsilon^2(1)} - \frac{\lambda_i}{2\phi_i \tilde{\sigma}_{\theta,i}^2} \right\}.$$

The perceived posterior variance after the first signal is

$$\tilde{\sigma}_{\theta,i|s(1)}^2 = (1 - \beta_i^*(1)) \tilde{\sigma}_{\theta,i}^2.$$

At each step, the optimization problem is identical except that the prior variance is replaced by the updated variance. Hence, the weight on the k th signal is

$$\beta_i^*(k) = \max \left\{ 0, \frac{\tilde{\sigma}_{\theta,i|s(k-1)}^2}{\tilde{\sigma}_{\theta,i|s(k-1)}^2 + \sigma_\varepsilon^2(k)} - \frac{\lambda_i}{2\phi_i} \cdot \frac{1}{\tilde{\sigma}_{\theta,i|s(k-1)}^2} \right\}, \quad (\text{E.2})$$

with the recursion

$$\tilde{\sigma}_{\theta,i|s(k)}^2 = \begin{cases} \frac{\sigma_{\theta,i}^2}{\omega_i}, & k = 0, \\ (1 - \beta_i^*(k)) \tilde{\sigma}_{\theta,i|s(k-1)}^2, & k \geq 1. \end{cases} \quad (\text{E.3})$$

To show the robustness of the static model, it suffices to show that our predictions hold to the case of two signals ($k = 2$). From here on, suppose that (E.2) always has an interior solution. Also, let $\sigma_\theta \equiv \sigma_{\theta,i}^2$, $\sigma_{\varepsilon k} \equiv \sigma_\varepsilon^2(k)$, $\lambda \equiv \lambda_i$, $\phi \equiv \phi_i$, and $\omega \equiv \omega_i$ for notational simplicity.

Hypothesis 1 (multi-signal). *Increasing overprecision leads to a lower update rate after the second signal.*

Support. The stage-1 optimal weight is

$$\beta^*(1) = \frac{\sigma_\theta}{\sigma_\theta + \omega\sigma_{\varepsilon 1}} - \frac{\lambda\omega}{2\phi\sigma_\theta}.$$

Define

$$T(\omega) := \frac{N(\omega)\sigma_\theta}{\omega}, \quad N(\omega) := 1 - \beta^*(1),$$

where $T(\omega)$ is the perceived prior variance entering stage 2. Then the stage-2 optimal weight is

$$\beta^*(2) = \frac{T}{\underbrace{T + \sigma_{\varepsilon 2}}_{=:F(T)}} - \frac{\lambda}{\underbrace{2\phi T}_{=:S(T)}}.$$

By the chain rule,

$$\frac{\partial\beta^*(2)}{\partial\omega} = \frac{dF}{dT} \frac{dT}{d\omega} - \frac{dS}{dT} \frac{dT}{d\omega}.$$

These derivatives can be shown as:

$$\frac{dT}{d\omega} = \sigma_\theta \left(\frac{1}{\omega} \frac{dN}{d\omega} - \frac{N}{\omega^2} \right),$$

$$\frac{dN}{d\omega} = \frac{\sigma_\theta\sigma_{\varepsilon 1}}{(\sigma_\theta + \omega\sigma_{\varepsilon 1})^2} + \frac{\lambda}{2\phi\sigma_\theta},$$

$$\frac{dF}{dT} = \frac{\sigma_{\varepsilon 2}}{(T + \sigma_{\varepsilon 2})^2},$$

$$\frac{dS}{dT} = -\frac{\lambda}{2\phi} \frac{1}{T^2}.$$

Substituting these yields

$$\frac{\partial\beta^*(2)}{\partial\omega} = \left[\frac{\sigma_{\varepsilon 2}}{(T + \sigma_{\varepsilon 2})^2} + \frac{\lambda}{2\phi} \frac{1}{T^2} \right] \sigma_\theta \left(\frac{1}{\omega} \left[\frac{\sigma_\theta\sigma_{\varepsilon 1}}{(\sigma_\theta + \omega\sigma_{\varepsilon 1})^2} + \frac{\lambda}{2\phi\sigma_\theta} \right] - \frac{N}{\omega^2} \right) \quad (\text{E.4})$$

The first term in brackets is positive. The second term in brackets is exactly equal to

$$\frac{d}{d\omega} \left(\frac{N}{\omega} \right) \quad \text{with} \quad \frac{N}{\omega} = \frac{\sigma_{\varepsilon 1}}{\sigma_\theta + \omega\sigma_{\varepsilon 1}} + \frac{\lambda}{2\phi\sigma_\theta}.$$

This derivative is negative. So the term in equation (E.4) is negative, which is qualitatively identical to the prediction of the static model.

Hypothesis 2 (multi-signal). *Increasing attention cost leads to a lower update rate after the second signal.*

Support. Again, the stage-1 optimal weight is

$$\beta^*(1) = \frac{\sigma_\theta}{\sigma_\theta + \omega \sigma_{\varepsilon 1}} - \frac{\lambda \omega}{2\phi \sigma_\theta}.$$

The perceived prior variance entering stage 2 is

$$T(\lambda) := \frac{N(\lambda) \sigma_\theta}{\omega}, \quad N(\lambda) := 1 - \beta_1^*.$$

The stage-2 optimal weight is

$$\beta^*(2) = \underbrace{\frac{T}{T + \sigma_{\varepsilon 2}}}_{=: F(T)} - \underbrace{\frac{\lambda}{2\phi T}}_{=: S(T)}.$$

First, note that N depends linearly on λ , hence

$$\frac{\partial N}{\partial \lambda} = -\frac{\partial \beta_1^*}{\partial \lambda} = \frac{\omega}{2\phi \sigma_\theta} \implies \frac{\partial T}{\partial \lambda} = \frac{\sigma_\theta}{\omega} \frac{\partial N}{\partial \lambda} = \frac{1}{2\phi}.$$

Using the chain rule,

$$\frac{\partial \beta^*(2)}{\partial \lambda} = \frac{dF}{dT} \frac{\partial T}{\partial \lambda} - \frac{\partial}{\partial \lambda} \left(\frac{\lambda}{2\phi T} \right),$$

and

$$\frac{dF}{dT} = \frac{\sigma_{\varepsilon 2}}{(T + \sigma_{\varepsilon 2})^2}.$$

Putting all these together,

$$\frac{\partial \beta^*(2)}{\partial \lambda} = \frac{1}{2\phi} \left[\frac{\sigma_{\varepsilon 2}}{(T + \sigma_{\varepsilon 2})^2} - \frac{1}{T} + \frac{\lambda}{2\phi} \frac{1}{T^2} \right] \quad (\text{E.5})$$

We need to evaluate the sign of the term inside the bracket. It is negative iff

$$\frac{\sigma_{\varepsilon 2}}{(T + \sigma_{\varepsilon 2})^2} < \frac{1}{T} - \frac{\lambda}{2\phi} \frac{1}{T^2}.$$

Equivalently,

$$\frac{\lambda}{2\phi} < T - \frac{\sigma_{\epsilon 2} T^2}{(T + \sigma_{\epsilon 2})^2}.$$

Since $(T + \sigma_{\epsilon 2})^2 \geq 4T\sigma_{\epsilon 2}$ due to the inequality of arithmetic and geometric means (AM-GM inequality), we can re-arrange the inequality above as:²³

$$\frac{\lambda}{2\phi} < T - \frac{T}{4} = \frac{3}{4}T. \quad (\text{E.6})$$

This inequality gives a conservative condition for verifying if $\frac{\partial \beta^*(2)}{\partial \lambda}$ is negative. Furthermore, we assume that the second-stage solution is interior ($\beta^*(2) > 0$). So it must be that

$$\frac{T}{T + \sigma_{\epsilon 2}} - \frac{\lambda}{2\phi T} > 0$$

This implies

$$\frac{\lambda}{2\phi} < \frac{T^2}{T + \sigma_{\epsilon 2}} = T - \frac{T\sigma_{\epsilon 2}}{T + \sigma_{\epsilon 2}} < T. \quad (\text{E.7})$$

Overall, conditions (E.6) and (E.7) imply that for interior solutions and plausible parameters, the derivative $\frac{\partial \beta^*(2)}{\partial \lambda}$ is negative, showing the robustness of the prediction from the static model.

Hypothesis 3 (multi-signal). *Overprecision amplifies the negative effects of attention costs on the update rate.*

Support. This hypothesis concerns the cross-derivative

$$\frac{\partial}{\partial \omega} \left(\frac{\partial \beta^*(2)}{\partial \lambda} \right).$$

Using equation (E.5), we can show the cross-derivative as:

$$\frac{\partial}{\partial \omega} \left(\frac{\partial \beta^*(2)}{\partial \lambda} \right) = \frac{1}{2\phi} \frac{\partial}{\partial T} \underbrace{\left[\frac{\sigma_{\epsilon 2}}{(T + \sigma_{\epsilon 2})^2} - \frac{1}{T} + \frac{\lambda}{2\phi} \frac{1}{T^2} \right]}_{=: \mathcal{H}(T, \lambda)} \cdot \frac{\partial T}{\partial \omega}. \quad (\text{E.8})$$

²³For any $a, b > 0$, AM-GM states that $\frac{a+b}{2} \geq \sqrt{ab} \Rightarrow (a+b)^2 \geq 4ab$.

From earlier, we know that $\frac{\partial T}{\partial \omega} < 0$. The sign of this term is therefore up to the partial derivative of $\mathcal{H}(T, \lambda)$ with respect to T :

$$\frac{\partial \mathcal{H}(T, \lambda)}{\partial T} = -\frac{2\sigma_{\varepsilon 2}}{(T + \sigma_{\varepsilon 2})^3} + \frac{1}{T^2} - \frac{\lambda}{\phi} \frac{1}{T^3}.$$

Let $u := \frac{T}{T + \sigma_{\varepsilon 2}}$. Then

$$\frac{2\sigma_{\varepsilon 2}}{(T + \sigma_{\varepsilon 2})^3} = \frac{2(1-u)}{T^2} u^2,$$

so

$$\frac{\partial \mathcal{H}}{\partial T} = \frac{1}{T^2} (1 - 2(1-u)u^2) - \frac{\lambda}{\phi T^3}.$$

The term inside the bracket is positive because the function $m(u) := 2(1-u)u^2$ attains its maximum $8/27$ at $u = 2/3$ and $1 - m(2/3) = 19/27$. Hence

$$\frac{\partial \mathcal{H}}{\partial T} \geq \frac{1}{T^2} \left(\frac{19}{27} - \frac{\lambda}{\phi T} \right).$$

Hence, $\frac{\partial \mathcal{H}}{\partial T} > 0$ whenever $\lambda/(\phi T) < 19/27$. Replacing T with the model's parameters gives the following condition for the robustness of hypothesis 3:

$$\text{if } \frac{\lambda}{\phi} \frac{\omega}{(1 - \beta^*(1))\sigma_{\theta}} < \frac{19}{27}, \quad \Rightarrow \quad \frac{\partial}{\partial \omega} \left(\frac{\partial \beta^*(2)}{\partial \lambda} \right) < 0,$$

i.e., overprecision amplifies the negative effect of attention costs on the second-signal update rate. Without the condition $\lambda/(\phi T) < 19/27$, however, the sign of the cross-derivative remains ambiguous.

D.2 Alternative cost functions

This section shows that the qualitative predictions of the model are robust beyond the Shannon specification of the cost function.

The optimization problem of the decision-maker has the general form of

$$\max_{\sigma_{\psi,i}^{-2} > 0} \underbrace{-\phi_i \text{Var}(\theta \mid s_i) - C(\sigma_{\psi,i}^{-2})}_{:= V(\sigma_{\psi,i}^{-2})}.$$

For ease of notation, let $a \equiv \tilde{\sigma}_{\theta,i}^2 = \frac{\sigma_{\theta,i}^2}{\omega_i}$, $q \equiv \sigma_{\psi,i}^{-2}$, $\lambda = \lambda_i$ and $\phi = \phi_i$. Standard normal-

normal algebra gives the posterior variance as a function of precision q :

$$\text{Var}(\theta \mid s_i) = \frac{a(\sigma_\epsilon^2 + q^{-1})}{a + \sigma_\epsilon^2 + q^{-1}}$$

Consider the generic convex cost function of $C(q) = \lambda G(q)$ with $G' > 0$ and $G'' \geq 0$.

The first-order condition for an interior optimum is

$$\frac{\partial V}{\partial q} = \phi \underbrace{\frac{a^2}{q^2 (a + \sigma_\epsilon^2 + q^{-1})^2}}_{:=H(q,a)} - \lambda G'(q) = 0 \quad (\text{E.9})$$

where $H(q, a)$ refers to the marginal benefit of attention. Thus, the optimal precision q^* is defined implicitly as the solution q to the function:

$$F(q, a, \lambda) = \lambda G'(q) - H(q, a).$$

We can use the implicit function theorem (IFT) to re-compute the hypotheses of the main model. In particular, IFT states:

$$\frac{\partial q^*}{\partial x} \equiv q_x^* = -\frac{F_x(q^*, x)}{F_q(q^*, x)}$$

where F_x refers to the partial derivative of F with respect to a variable x . To compute our hypotheses, x is either λ or a . This gives the following numerators

$$F_a = -H_a = -\frac{2\phi a(\sigma_\epsilon^2 + q^{-1})}{q^2(a + \sigma_\epsilon^2 + q^{-1})^3} < 0, \quad \text{and} \quad F_\lambda = G'(q) > 0.$$

The denominator is equal to

$$F_q = \lambda_i \underbrace{G''(q_i)}_{\geq 0} - \underbrace{H_q(q, a)}_{< 0} > 0.$$

Jointly, they give

$$q_\lambda^* = -\frac{F_\lambda}{F_q} < 0 \quad \text{and} \quad q_a^* = -\frac{F_a}{F_q} > 0.$$

Since

$$\frac{\partial \beta}{\partial x} = \frac{\partial \beta}{\partial q^*} \frac{\partial q^*}{\partial x} = \beta_q(q^*, a) q_x^*$$

and

$$\beta_q(q^*, a) = \frac{a}{q^2(a + \sigma_\epsilon^2 + q^{-1})^2} > 0,$$

we can confirm that the first two hypotheses of the main model extend to this generic cost function: Attention cost is negatively associated with the optimal update ($\beta_\lambda < 0$) and prior variance is positively associated with the optimal update ($\beta_a > 0$). For hypothesis 3, we need to show the cross derivative:

$$\frac{\partial}{\partial \omega} \left(\frac{\partial \beta^*}{\partial \lambda} \right) = \frac{\partial a}{\partial \omega} \frac{\partial}{\partial a} \left(\frac{\partial \beta^*}{\partial \lambda} \right) < 0.$$

Since $\partial a / \partial \omega < 0$ by construction, it suffices to show the remaining term is positive to show the robustness. This term is equal to

$$\frac{\partial}{\partial a} \left(\frac{\partial \beta^*}{\partial \lambda} \right) = \beta_{qa} q_\lambda + \beta_q q_{\lambda a}.$$

We have already established that $q_\lambda < 0$ and $\beta_q^* > 0$. The remaining terms have the following signs:

$$\beta_{qa}^* = \frac{\sigma_\epsilon^2 + q^{-1} - a}{q^2(a + \sigma_\epsilon^2 + q^{-1})^3}$$

and

$$q_{\lambda a} = -\frac{G''(q)q_a}{F_q} + \frac{F_\lambda}{F_q^2} \left(\underbrace{F_{qa}}_{-H_{qa}(q^*, a) > 0} + \underbrace{F_{qq}}_{\lambda G'''(q^*) - H_{qq}(q^*, a)} q_a \right).$$

The former is negative whenever $a > \sigma_\epsilon^2 + q^{-1}$ and the latter is negative for relatively small $\lambda G''' - H_{qq}$, which holds for many empirically relevant cost functions (linear, power, exponential).

Overall, the model predictions remain robust for a large range of values for the main parameters as long as we have convex increasing cost functions.